

# ARCHANA KUMARI

## CURRICULUM VITAE

Full-Stack Software Engineer · Python · Django · FastAPI · React · Next.js

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## CURRICULUM VITAE PROFILE & ENGINEERING PHILOSOPHY

**Full-Stack Software Engineer** with 2+ years of production experience architecting high-throughput, fault-tolerant web applications and distributed architectures. Demonstrated excellence in building enterprise platforms from scratch, including end-to-end **E-Commerce management, real-time Inventory Management across 1,000+ SKUs, and retail POS billing systems**. Deep expertise in backend asynchronous job processing (Celery & RabbitMQ pipelines processing 10,000+ daily jobs with 0% data loss), sub-120ms API latency optimization via Redis caching and PostgreSQL query indexing, multi-tenant SaaS schema isolation guaranteeing zero cross-tenant leakage, payment gateways (PhonePe), logistics tracking APIs (Ekart), and AI integration with the OpenAI API.

## COMPREHENSIVE TECHNICAL COMPETENCIES

|                                   |                                                                                                                     |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Backend Architecture:</b>      | Python 3.x, Django, Django REST Framework, FastAPI, Node.js, Express.js, RESTful API Design                         |
| <b>Asynchronous &amp; Queues:</b> | Celery Distributed Task Queue, RabbitMQ Message Broker, Redis In-Memory Store & Cache, Pub/Sub                      |
| <b>Frontend Engineering:</b>      | React, Next.js (App Router/SSR), Redux Toolkit, RTK Query, JavaScript (ES6+), TypeScript, Tailwind CSS, Formik, Yup |
| <b>Databases &amp; Storage:</b>   | PostgreSQL (Indexing, Constraints, REVOKE), MySQL, MongoDB, Firebase, Supabase, Cloudinary                          |
| <b>System Architecture:</b>       | Microservices, Event-Driven Architecture, Multi-Tenant SaaS Partitioning, Append-Only Ledgers, RBAC, JWT Auth       |
| <b>DevOps &amp; Tooling:</b>      | Docker, Docker Compose, Nginx Reverse Proxy, AWS (EC2, S3), GitHub Actions CI/CD, Git, Linux/Bash                   |
| <b>Third-Party &amp; AI:</b>      | OpenAI API (Prompt Engineering, LLM Integration), PhonePe Payment Gateway, Ekart Logistics API                      |

## PROFESSIONAL WORK EXPERIENCE

**Full Stack Developer · Cloud Software Solution** · Bangalore, India Nov 2025 – Present

- E-Commerce, Inventory & POS (Engineered from Scratch):** Architected and built completely from scratch an enterprise E-Commerce management platform, real-time Inventory Management system, and POS billing engine (Django, Next.js) syncing live stock across **1,000+ retail SKUs** with automated order-to-dispatch workflows.
- High-Throughput Asynchronous Task Pipelines:** Designed Celery + RabbitMQ distributed worker pipelines processing **10,000+ daily background tasks** (order notifications, stock updates, invoice generation, reconciliation) with **zero message loss**.
- Latency Optimization & Database Performance:** Slashed API response times by **40% (sub-120ms P95)** by designing a multi-layer Redis caching strategy and optimizing PostgreSQL compound indexing and query execution plans.
- Franchise RBAC & Multi-Location Security:** Built modular Franchise Management module and granular Role-Based Access Control (RBAC) ensuring secure, role-specific data access across multi-location franchise staff and store managers.
- Payment & Logistics Integrations:** Integrated PhonePe payment gateway with idempotent webhook validation and automated reconciliation; integrated Ekart API for real-time parcel dispatch tracking and delivery visibility.
- Automated CI/CD Pipelines:** Configured automated CI/CD pipelines via GitHub Actions and Docker, achieving consistent zero-downtime releases and automated containerized testing.

**Software Engineer · Comestro Pvt Ltd** · India Sept 2024 – Oct 2025

- Online Examination Engine:** Spearheaded full-cycle development of an online examination platform (React + RTK Query) serving **5,000+ concurrent students** across 3 adaptive difficulty tiers with anti-cheating mechanisms and auto-submission timers.
- Multi-Tenant SaaS Architecture:** Architected multi-tenant SaaS platform enforcing strict schema isolation, guaranteeing **100% data partition integrity with zero cross-tenant leakage**.
- Dynamic Form Architecture:** Built 15+ complex dynamic forms with Formik + Yup schema validation, reducing form submission validation errors and support tickets by **35%**.
- Client Performance & Security:** Implemented JWT authentication with role-based route guards; cut redundant API roundtrips by **50%** through RTK Query normalized caching.

**Full Stack Intern · Techjet.ai** · India March 2024 – Aug 2024

- Built secure full-stack authentication flows (protected routes, sessions, JWT) across a React + Django platform.
- Integrated 10+ RESTful APIs and managed centralized application state using Redux Toolkit, decreasing code duplication by **40%**.

MAJOR TECHNICAL PROJECTS & SYSTEM ARCHITECTURE

AI Resume Screener & Candidate Intelligence Platform · [github.com/Archana-PC/docsense-ai](#)

- Engineered a 4-service microservices application that automatically parses PDF resumes, extracts competencies, and ranks candidates against job descriptions using the OpenAI API.
- Architected asynchronous processing pipeline (PDF parser → Celery worker + Redis queue → AI scorer → PostgreSQL) preventing thread blocking during heavy document processing.
- Crafted optimized prompt engineering schemas for deterministic scoring matrices and structured JSON outputs.
- Containerized the full stack with Docker Compose and healthcheck probes for reliable single-command deployment.

Django, FastAPI, React, Celery, Redis, OpenAI API, PostgreSQL, Docker

Brahmo Data Integrity — Append-Only Knowledge Graph · [github.com/Archana-PC/brahmo-data-integrity](#)

- Developed an append-only medical knowledge graph enforcing PostgreSQL database-level REVOKE permissions to guarantee tamper-proof audit trails against unauthorized UPDATE/DELETE operations.
- Engineered temporal SUPERSEDE node relationships for clinical history preservation, ensuring historical state reconstruction without data mutation.
- Built an interactive React/Vite graph explorer interface querying complex relational nodes with sub-50ms latency.
- Implemented verification test suites validating rollback resilience and unauthorized mutation prevention.

Python 3.11, FastAPI, Supabase, PostgreSQL, React, Vite

AuraCommerce — Enterprise E-Commerce, POS & Inventory Platform · [github.com/Archana-PC](#)

- Architected and built completely from scratch an end-to-end commerce system managing live stock across 1,000+ retail SKUs and multi-location franchises.
- Implemented automated order-to-dispatch workflows with idempotent PhonePe payment webhooks and live Ekart tracking APIs.
- Designed multi-layer Redis caching strategy slashing catalog and checkout API response times by 40% (sub-120ms P95).

Django, Next.js, Celery, RabbitMQ, Redis, PostgreSQL, PhonePe, Ekart API

StreamCore — Video Streaming & Transcoding Backend · [github.com/Archana-PC/backend\\_javascript](#)

- Architected RESTful microservices for chunked video ingestion, watch history tracking, and user channel subscriptions.
- Engineered complex MongoDB aggregation pipelines for high-throughput pagination, search, and recommendation feeds.
- Secured API endpoints with HMAC SHA-256 JWT refresh/access token rotation and bcrypt password hashing.

Node.js, Express.js, MongoDB, Mongoose, JWT Auth, Cloudinary, Multer

EDUCATION & ACADEMIC CREDENTIALS

Master of Computer Applications (MCA) · Presidency College, Bangalore

2021 – 2022

Core Focus: Distributed Systems, Advanced Database Management Systems, Object-Oriented Software Design, Cloud Architecture, Enterprise Web Applications.

Bachelor of Computer Applications (BCA) · Purnea College, Purnea

2016 – 2020

Core Focus: Data Structures & Algorithms, Relational Database Design, Operating Systems, Computer Networks, Systems Programming.

ENGINEERING METHODOLOGIES & CORE STANDARDS

Architecture: Microservices, Event-Driven, Distributed Queues, Multi-Tenant SaaS

Quality & Testing: Unit/Integration Testing, CI/CD GitHub Actions, Dockerized Environments

Security & Auth: JWT Token Rotation, Role-Based Access Control (RBAC), Idempotency

Data Integrity: Append-Only Auditing, ACID Transactions, Compound Indexing

Performance: Sub-120ms P95 Latency, Redis Layered Caching, RTK Query Normalization

Collaboration: Agile / Scrum, Code Reviews, Git Flow, API Contract Documentation